

Project Execution & Working Demonstration

1. Azure Portal — proves the Azure networking infrastructure and VPN connection are running

2. Localhost Guardian Dashboard — proves your diagnostic application is running and reading the real VPN state

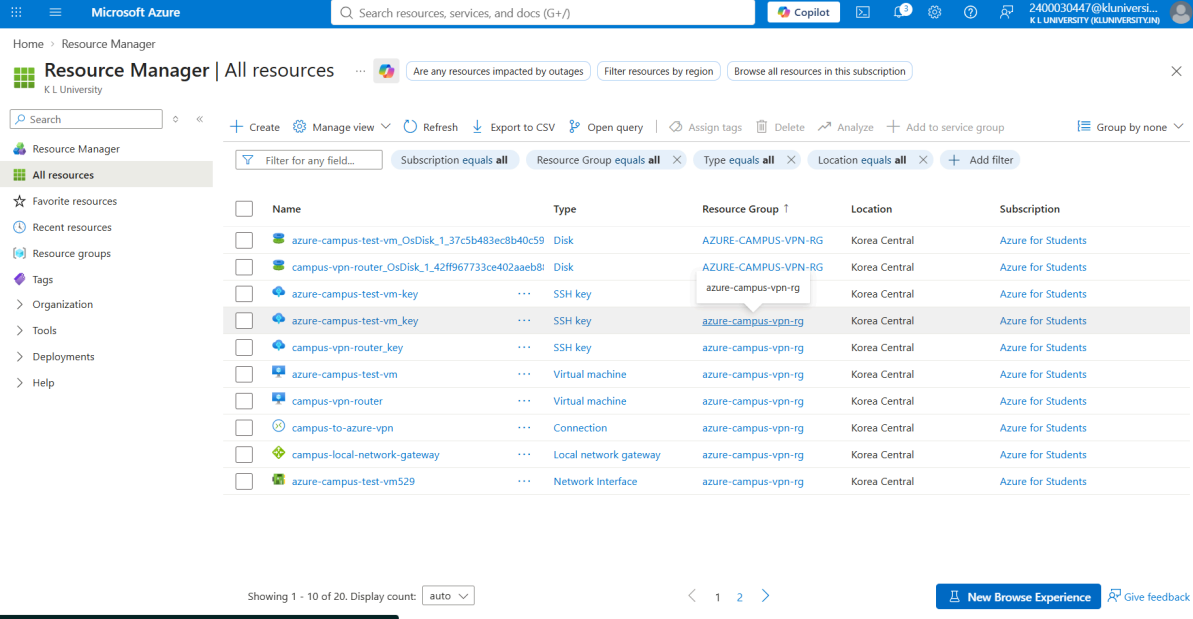
1. Azure Portal – Project Infrastructure

Start by opening the Azure Portal and navigating to:

Resource Groups → **azure-campus-vpn-rg**

Take a screenshot showing the resources:

- **azure-campus-vpn-gateway**



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is titled 'Resource Manager | All resources'. On the left, there is a sidebar with navigation options: 'All resources', 'Favorite resources', 'Recent resources', 'Resource groups', 'Tags', 'Organization', 'Tools', 'Deployments', and 'Help'. The main area displays a table of resources. The table has columns for 'Name', 'Type', 'Resource Group', 'Location', and 'Subscription'. The resources listed are:

Name	Type	Resource Group	Location	Subscription
azure-campus-test-vm_OsDisk_1_37c5b483ec8b40c59	Disk	AZURE-CAMPUS-VPN-RG	Korea Central	Azure for Students
campus-vpn-router_OsDisk_1_42ff967733ce402aeb8f	Disk	AZURE-CAMPUS-VPN-RG	Korea Central	Azure for Students
azure-campus-test-vm-key	SSH key	azure-campus-vpn-rg	Korea Central	Azure for Students
azure-campus-test-vm_key	SSH key	azure-campus-vpn-rg	Korea Central	Azure for Students
campus-vpn-router_key	SSH key	azure-campus-vpn-rg	Korea Central	Azure for Students
azure-campus-test-vm	Virtual machine	azure-campus-vpn-rg	Korea Central	Azure for Students
campus-vpn-router	Virtual machine	azure-campus-vpn-rg	Korea Central	Azure for Students
campus-to-azure-vpn	Connection	azure-campus-vpn-rg	Korea Central	Azure for Students
campus-local-network-gateway	Local network gateway	azure-campus-vpn-rg	Korea Central	Azure for Students
azure-campus-test-vm529	Network interface	azure-campus-vpn-rg	Korea Central	Azure for Students

At the bottom of the page, there is a footer with the URL: <https://portal.azure.com/#resource/subscriptions/2c72be0e-e1da-4875-8810-124fb2867...>

Azure resources deployed for the campus-to-Azure Site-to-Site VPN

2. Azure Virtual Network

Open:

azure-campus-vnet → **Overview**

services, and docs (G+/)

Copilot

2400030447@kluniver...
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azure-campus-vpn-gateway

Virtual network gateway

Search

Refresh Move Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Configuration
 - Connections
 - Point-to-site configuration
 - NAT Rules
 - Maintenance
 - Properties
 - Locks
- Monitoring
- Automation
- Help

Essentials JSON View

Resource group (move)	: azure-campus-vpn-rg
Location	: Korea Central
Subscription (move)	: Azure for Students
Subscription ID	: 2c72be0e-e1da-4875-8810-124fb28b7...
SKU	: VpnGw2AZ
Gateway type	: VPN
VPN type	: Route-based
Virtual network	: azure-campus-vnet
Public IP address	: 4.217.131.246 (azure-campus-vpn-pip)
Tags (edit)	: Add tags

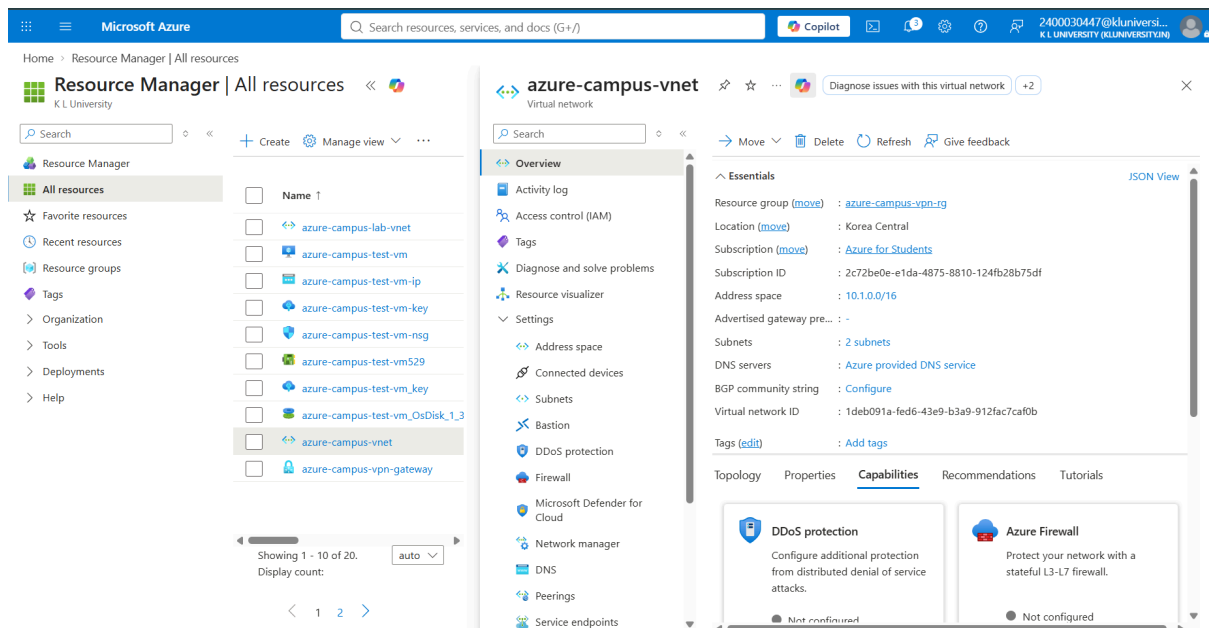
Health check
Perform a quick health check to detect possible gateway issues
[Go to Resource health](#)

Advisor Recommendations
Check Critical, Warning, and Informational Recommendations
[Go to Advisor](#)

Show data for last **1 hour** 6 hours 12 hours 1 day 7 days 30 days

Total tunnel ingress

Total tunnel egress

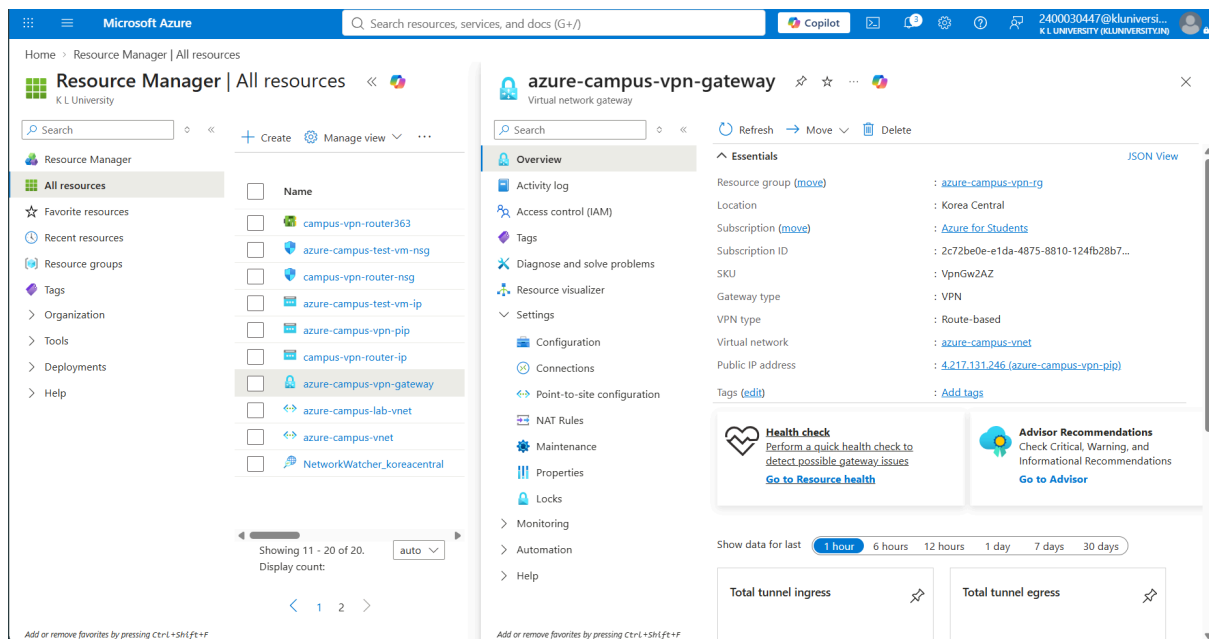


Azure Virtual Network and subnet configuration

3. Azure VPN Gateway

Open:

azure-campus-vpn-gateway



Azure VPN Gateway configuration

4.Site-to-Site VPN Connection

Open:

campus-to-azure-vpn

The screenshot shows the Microsoft Azure portal interface. The main view is the 'Connections' page for the 'azure-campus-vpn-gateway'. The table below shows a single connection:

Name	Status	Connection type	Peer
campus-to-azure-vpn	Connected	Site-to-site (IPsec)	campus-local-netwo...

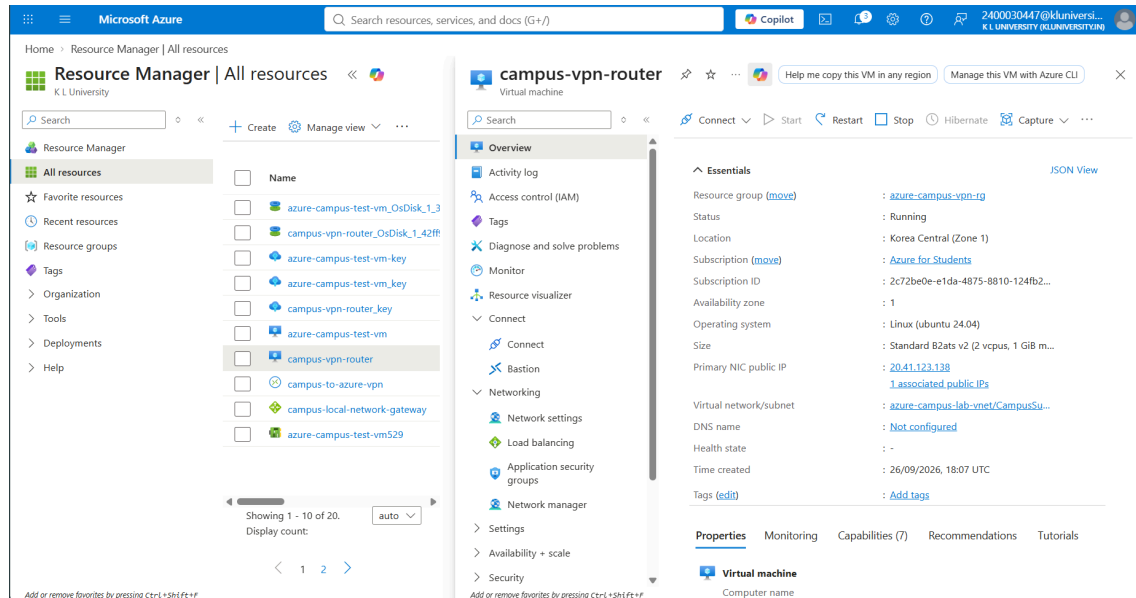
The left sidebar shows the 'All resources' list with 'azure-campus-vpn-gateway' selected. The right sidebar shows the 'Connections' menu with options like 'Point-to-site configuration', 'NAT Rules', 'Maintenance', 'Properties', 'Locks', 'Monitoring', 'Automation', and 'Help'.

Site-to-Site VPN connection established between the simulated campus network and Azure

5. Campus VPN Router

Open:

campus-vpn-router



6. Show Actual IKE/IPsec Negotiation

SSH into the campus router.

Run:

sudo ipsec statusall

```
azureuser@campus-vpn-router:~$ sudo ipsec statusall
Status of IKE charon daemon (strongSwan 5.9.13, Linux 6.17.0-1022-azure, x86_64):
  uptime: 73 minutes, since Sep 26 19:32:29 2026
  malloc: sbrk 2887680, mmap 0, used 1188352, free 1699328
  worker threads: 11 of 16 idle, 5/0/0/0 working, job queue: 0/0/0/0, scheduled: 4
  loaded plugins: charon aesni aes rc2 sha2 sha1 md5 mgf1 random nonce x509 revocation constraints pubkey pkcs1 pkcs7 pk
cs12 pgp dnskey sshkey pem openssl pkcs8 fips-prf gmp agent xcbc hmac kdf gcm drbg attr kernel-netlink resolve socket-de
fault connmark stroke updown eap-mschapv2 xauth-generic counters
Listening IP addresses:
  10.20.0.4
Connections:
  azure-s2s: %any...4.217.131.246 IKEv2, dpddelay=30s
  azure-s2s: local: [20.41.123.138] uses pre-shared key authentication
  azure-s2s: remote: [4.217.131.246] uses pre-shared key authentication
  azure-s2s: child: 10.20.0.0/24 === 10.1.0.0/16 TUNNEL, dpdaction=start
Security Associations (1 up, 0 connecting):
  azure-s2s[8]: ESTABLISHED 47 minutes ago, 10.20.0.4[20.41.123.138]...4.217.131.246[4.217.131.246]
  azure-s2s[8]: IKEv2 SPIs: 732ffe33c7075a1d_i c13f0f71d2fe20b2_r*, pre-shared key reauthentication in 119 minutes
  azure-s2s[8]: IKE proposal: AES_CBC_256/HMAC_SHA2_256_128/PRF_HMAC_SHA2_256/MODP_1024
  azure-s2s[5]: INSTALLED, TUNNEL, reqid 1, ESP in UDP SPIs: c20669ad_i a762a142_o
  azure-s2s[5]: AES_GCM_16_256, 0 bytes_i, 0 bytes_o, rekeying in 42 minutes
  azure-s2s[5]: 10.20.0.0/24 === 10.1.0.0/16
```

7. Prove REVERSE Connectivity

```
azureuser@campus-vpn-router:~/campusvpn-guardian$ ping -c 4 10.1.0.4
PING 10.1.0.4 (10.1.0.4) 56(84) bytes of data:
64 bytes from 10.1.0.4: icmp_seq=1 ttl=64 time=1.58 ms
64 bytes from 10.1.0.4: icmp_seq=2 ttl=64 time=1.62 ms
64 bytes from 10.1.0.4: icmp_seq=3 ttl=64 time=1.51 ms
64 bytes from 10.1.0.4: icmp_seq=4 ttl=64 time=2.53 ms

--- 10.1.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.507/1.807/2.529/0.418 ms
```

8. Start CampusVPN Guardian

SSH into the campus router:

cd ~/campusvpn-guardian

python3 app.py

```
azureuser@campus-vpn-router:~$ cd ~/campusvpn-guardian
azureuser@campus-vpn-router:~/campusvpn-guardian$ python3 app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.20.0.4:5000
Press CTRL+C to quit
127.0.0.1 - - [26/Sep/2026 20:46:46] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Sep/2026 20:46:53] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Sep/2026 20:46:55] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Sep/2026 20:46:56] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Sep/2026 20:46:57] "GET / HTTP/1.1" 200 -
```

9. Create the Localhost Tunnel

command:

& "C:\Windows\System32\OpenSSH\ssh.exe" -i

"C:\Users\Asus\Downloads\campus-vpn-router_key.pem" -L
5000:127.0.0.1:5000 azureuser@20.41.123.138

Then open Chrome:

http://localhost:5000

